Erik Hogset

erikrh@bu.edu | +1 (814) 310 1649 | Boston, MA | linkedin.com/in/erikhogset |
Portfolio: erikhogset.github.io

Education

Boston University College of Engineering, Boston, MA — Expected May 2026

B.S. in Mechanical Engineering

Relevant Coursework: Manufacturing Processes, Automation & Manufacturing Methods, Additive Manufacturing, Electromechanical Design, Mechanics of Materials, Senior Design

Work Experience

Starbucks, Breezewood, PA and Boston, MA — Jun. 2021 – Sep. 2025

Barista

Balanced part-time employment with full-time Mechanical Engineering coursework
Worked in a fast-paced, team-based environment requiring accuracy, multitasking, and efficiency
Maintained consistent performance and reliability over multiple years of employment

Engineering Product Innovation Center (E.P.I.C.), Boston, MA. — May 2021 - Present

Laboratory Assistant

Supported prototyping and fabrication for student and faculty projects in a high-traffic makerspace
Operated CNC mills, CNC lathes, manual mills and lathes, and 3D printers (*FFF* and *SLA*)
Generated CAM toolpaths and G-code using SolidWorks and Autodesk Fusion 360
Assisted with part fixturing, tolerancing, and design adjustments for manufacturability
Enforced lab safety and proper machine operation while supporting multiple users

Projects

Adaptive LED Lamp – Introduction to Engineering Design (*EK210*)

- Designed and built a motion-activated LED lamp using electronic control and physical prototyping
- Implemented ambient-light and time sensing to limit blue light and adjust brightness for accessibility

Electromechanical Interactive Ouija Board – Electromechanical Design (*ME360*)

- Built a 2-axis, belt-driven motion system to position a planchette using AI-generated responses
- Integrated stepper motors and motion control to execute precise planar positioning
- Developed code to retrieve AI responses from voice input and convert responses to coordinate targets

Compact Fishing Kit Product – Automation & Manufacturing Methods (*ME345*)

- Designed a consumer product and developed a CAD-to-CAM workflow for CNC manufacturing
- Created CNC toolpaths and developed robot programs to tend machines and assemble parts
- Built an automated production line coordinating conveyors, robot programs, and CNC operations

Reactive Road Sign System – Senior Design I & II (*ME460/461*)

- Designed a retrofit mechanical system for roadside signposts to deflect under extreme wind loads
- Developed and evaluated joint concepts using CAD, materials selection, and failure analysis
- Evaluated manufacturability, corrosion resistance, and geometry-driven failure modes
- Produced physical prototypes and formal design documentation for engineering design reviews

Skills

Manufacturing: Manual lathe, manual milling machine, CNC milling machine, CNC lathe, 3D printing (FFF and SLA), and TIG welding

CAD Skills: Onshape, SolidWorks, Autodesk Fusion 360,

Coding Skills: MATLAB, Python, C, and Arduino IDE